

Managing partner health means knowing what healthy is.

Problem framing

healthy = maximize (**order value** – **value lost** – **cost of capture**)

NORTH STAR METRIC: **value lost per \$ presented**

THE JOB

Order value is the payment stack's **input** — the stack cannot increase it, only deliver it. The stack's whole job:

- **Performance** — lose less order value through the funnel.
- **Cost** — charge less for the passage.

OUT OF SEGMENT INSIGHTS

Minimizing value lost is a **pathing problem**: many routes to the same destination, each with its own quality and price, changing by the minute. CDNs and telecoms solved it long ago.

Both settled on the same rule: **quality decides the route, price only breaks ties between routes that pass the quality bar.**

WHY IT MATTERS

- **The scale** — every percentage point lost in the funnel is lost from the profit margin. Large clients feel it.
- **It's invisible** — value loss in this funnel doesn't arrive as an invoice or a cost line. It's an opportunity cost merchants never feel. Add multiple partners and geo coverage, and the problem compounds.
- **On your own** — as a merchant, you are on your own. As Primer, we have the data across segments to provide benchmarks.

NOTE — THE DECISION IS THE COST

Partner health is as much about finding the best configuration as about **who decides and implements the change**. In large merchants that enablement is the biggest cost of the decision — lowering it is the point of orchestration.

Approach

DIAGNOSIS

Traceability — every transaction's path through the payment flow: which partners touched it, at every level, and what each level did to the end margin. Primer's Observability already captures this per-transaction data sliced by dimension — the raw material is assumed to exist; traceability means connecting it per-path.

Cohorting — transactions grouped by behaviour through the funnel. Routing knowledge isn't just partner-by-partner — it's learning which kinds of transactions belong on which path.

Sampled comparison — transaction windows scored against history vs the same moment, on own data · industry segment · all of Primer's merchants.

PROPOSAL — FROM SCORE TO DECISION

At every level of the flow the merchant chooses between competing providers. Each provider gets a **score: its measured contribution to the health formula** — value delivered, minus the value it lost, minus what it charged — computed on the transaction windows from Diagnosis and compared three ways:

- **vs itself** — same provider, same conditions: history, seasonal baseline, same-moment windows.
- **vs peers** — same level, compared on like-for-like traffic.

- **vs all Primer merchants** — the network benchmark.

A score that dips below its own history or its peers is the trigger for everything that follows. Providers are scored only on what they control — non-addressable failures (insufficient funds, customer error) are excluded.

PRIORITIZATION

- ① **Knowledge.** Scores, traces, and cohort analysis fed into the AI Companion as data to power manual investigations — per transaction, and per sample window across many transactions. The humans investigating validate the implementation itself.
- ② **Alerting.** A notification system on top of step 1: score deviations trigger a short alert with the summary attached — getting ahead of incidents instead of finding them in the revenue numbers.
- ③ **Recommendation.** AI learns from human decisions and starts proposing its own **Change / Stay** — a reinforcement cycle, benchmarked on predicted vs realized outcomes. This gets surfaced in the Workflows module as recommended workflows, rules and paths.
- ④ **Automation.** Confirmation of the approach: unlocked per level as the model matches or beats a human operator's accuracy in less time.

Trade-offs

THE MAIN ONE: AUTOMATION STARVES ITS OWN SCORECARDS

Automation routes to preferred paths. Unused paths stop generating data, their scores go stale, and the system can't tell whether a rejected path improved — it optimises into a local maximum and calls it healthy.

Four defences, cheapest first:

- **Network data as free exploration.** A path this merchant abandoned still runs live for other Primer merchants — their traffic keeps its score fresh at no cost here.
- **Confidence decays without traffic.** Silence becomes uncertainty, and uncertainty triggers the next two.
- **An explore budget.** A small deliberate share of traffic on alternatives — a measurable cost of keeping options honest, spent in low-value segments first.
- **Probes where traffic can't go.** Liveness and latency checks keep a dormant path observable without sending real money through it.

OTHER TRADE-OFFS

- **Speed vs correctness** — automation is worth it only if it takes a good decision. Otherwise the human is slower BUT better.
- **Wrong attribution** — one confident wrong verdict costs more trust than ten right ones earn.
- **Flapping** — reacting fast enough to matter without oscillating traffic between providers.
- **Alert sensitivity** — a low threshold catches small real dips but also fires on noise, and operators start ignoring alerts. A high threshold keeps every alert credible but lets slow erosion run unnoticed. Where the line sits is a decision, not a default.
- **Commercial commitments** — volume deals with acquirers constrain what "optimal" is allowed to mean.
- **Thin data** — small merchants and narrow corridors can't support their own scorecards; they lean on segment and network scopes.
- **Gaming** — once providers know they're scored, some will optimise the score rather than the outcome.

Success metrics

NORTH STAR

Value lost per \$ presented

The formula's own term — proves the product worked, not just ran.

CUSTOMER

Satisfaction score

How much did this actually help you?

QUALITY

% false-flag warnings

Discarded by the human.

SPEED

Time to decision — Change / Stay

System side — sample windows needed to detect a trend.

Merchant side — from warning to decision.

Assumptions and open questions

ASSUMED

- We can follow each transaction through every level of the flow.
- We know what every provider charges, well enough to calculate real costs.
- We're allowed to use data from all merchants for comparisons.
- We have enough past incidents to test the scoring against before showing it to anyone.
- Merchants have someone who will actually read the alerts and act.

OPEN

- When comparing a merchant to "similar merchants" — similar how? Industry, size, geography?
- When everything dips at once, how do we tell "this partner has a problem" from "the whole market has a problem"?
- Should partners see the scores we give them?
- How small can a merchant be before their own data is too thin to score honestly?
- Is this part of the product, or something merchants pay extra for?